Card Sort Report
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**Methods**

We recruited 10 participants from among our friends and family and each conducted five card sorts. Each of the card sorts were one on ones. Some were done in-person with physical cards, and some done virtually, over phone call, by dragging boxes on Google Drawing. We instructed the participants to sort the cards into categories that made sense to them and write the labels for each category. They shared thoughts as they sorted, and afterwards, we asked them questions of what confused them or why they chose what labels and categories they created.

**Trends in the Card Sort and Isolated Incidents**

Three types of labels that showed up in nearly every sort were some variation of products, tools, and information. This general pattern was what we expected to find, since those were the three types of things we decided to include in our card sort. We were surprised by other findings, such as what sub-categories our participants included, and some cards that split between different categories.

Many cards had strong groups and were sorted together by most participants. Silk bonnet, satin scrunchies, and silk pillowcase were together in every sort. Similarly, every participant sorted the three shampoos together. The two influencers were always put in the same group, and frequently placed with other informational items. Curl cream and mousse were together in 90% of the sorts, as well as microfiber hair towel and towel bonnet.

One theme that may not show up in our standardized labels is how often our participants emphasized the sequential nature of a curly hair routine. Many people sorted into specific times you would use certain products (“wash day”, “curly hair aftercare”, “sleep”). One participant organized her piles step by step, based on what order the cards would show up in a routine. Another participant combined liquid products and physical tools into the same categories, because they would be used in the same steps of the routine.

Some participants did broader categories (Product, Tools, and Resources) while others did more specific categories that usually landed within the other participants' larger category (Product -> Shower routine). This leads us to believe that having a tree might be better, with the larger categories

as the trunk and smaller categories as the leaves. Some people had different branches with information. Some grouped people separately, and others combined them with instructions and tips.

The cards that were splitters were Homemade flax-seed hair gel recipe, “Plopping” instructions, and Leave-in conditioner. The recipe had the biggest split, with 50% putting it with products, 30% putting it in information, and 20% putting it in a whole new category (Recipes, Why not try it) because they didn’t feel it belonged. “Plopping” instructions 80% had it with instructions, and 20% had it with “wash day stuff” and “Keep Curls Fresh,” which have more product-related items. The last splitter was leave-in conditioner. Three people said during washing and two said after.

**Categories and Labels**

According to the results, it would make most sense to have two organizational schemes: by type (products, tools, education), and a sequential organization of when you would use the product or tool or need the information.

Here are the labels and sub-labels for the first organizational scheme, as well as an explanation of why we are including that category.

- Curly Hair Products:
    - 10/10  Every sort had some version of products. Some split it into styling and shampoo/conditioner. This is from 6’s label and is topical.
    - Shampoo/Conditioner:
        - 5/10 separated into washing and 2/10 explicitly said shampoo and conditioner this is from 1’s label and is topical.
    - Styling Products:
        - 4/10 separated based on styling products, taken from 4’s label and is topical
- Tools:
    - 9/10 Most sorted tools either together or between the following two groups. This label was taken from 3, 9, 10 all used “tools”. These are topical
    - Styling: 4/10
        - Physical objects to help define and refresh curls, label taken from 5, 9,10
    - Protecting

        - 5/10 Objects to put hair in for sleep or after shower to keep the curls looking nice. Label from 5, 10
- Education
    - 10/10 Everyone put all of the education materials together. Some broke it down into smaller categories. Label taken from 5, 8. These are topical
    - Instructions
        - 2/10 “How to” information. The two labels given were “to-do” (1) and “advise” (2) we combined them into instructions
    - Science
        - 4/10 Information on how the hair works, we used the label from 1 ,2, 9 who all used “science”
    - Influencers
        - 7/10 sorted them separately. These are social media people who post on curly hair.

Here are the labels for the sequential organization:

- Washing
    - 5/10 specifically stated wash day. Unlike the one above, this is sequential and includes not only products (liquid and plastic), but also educational material as well, including videos from influencers. Washing label from 3, 4
- Drying
    - 2/10 same with washing different kinds of items related to drying, label from 4
- Styling
    - 3/10 same as before with different items related to styling, label from 4, 5
- Sleeping
    - 3/10 same as before but related to sleeping, label from 4, 10
- Touch up
    - 2/10 same as before but related to how to keep your hair looking nice a few days after wash day. Label is from 1.

**What We’ll Ignore**

One thing we'll ignore is that most people who labeled our two influencer cards into their own group called it "people," and no one labeled it "influencers." We felt that "influencers" was a more clear label, and that our participants may not have used it because our cards said influencers on them.

For now, we're ignoring the input from Participant 4 to put recipes and hair cuts as their own categories. If we add more information on hair cuts or more recipes, we may make those a subcategory under education, but for now, they can go under Education -> Instructions since they each only have one item.

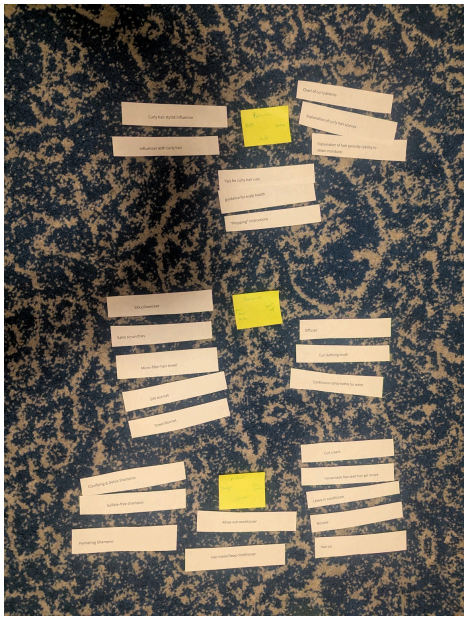
We'll also ignore how Participant 7 put the flaxseed hair gel in a "Shower routine" category, since it is actually a product you would use after the shower, not during.

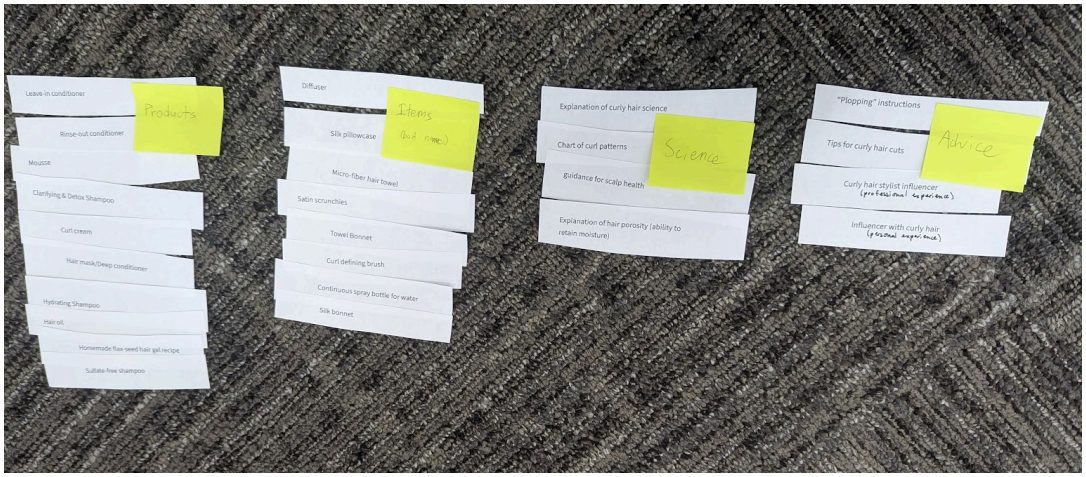
**Appendix**

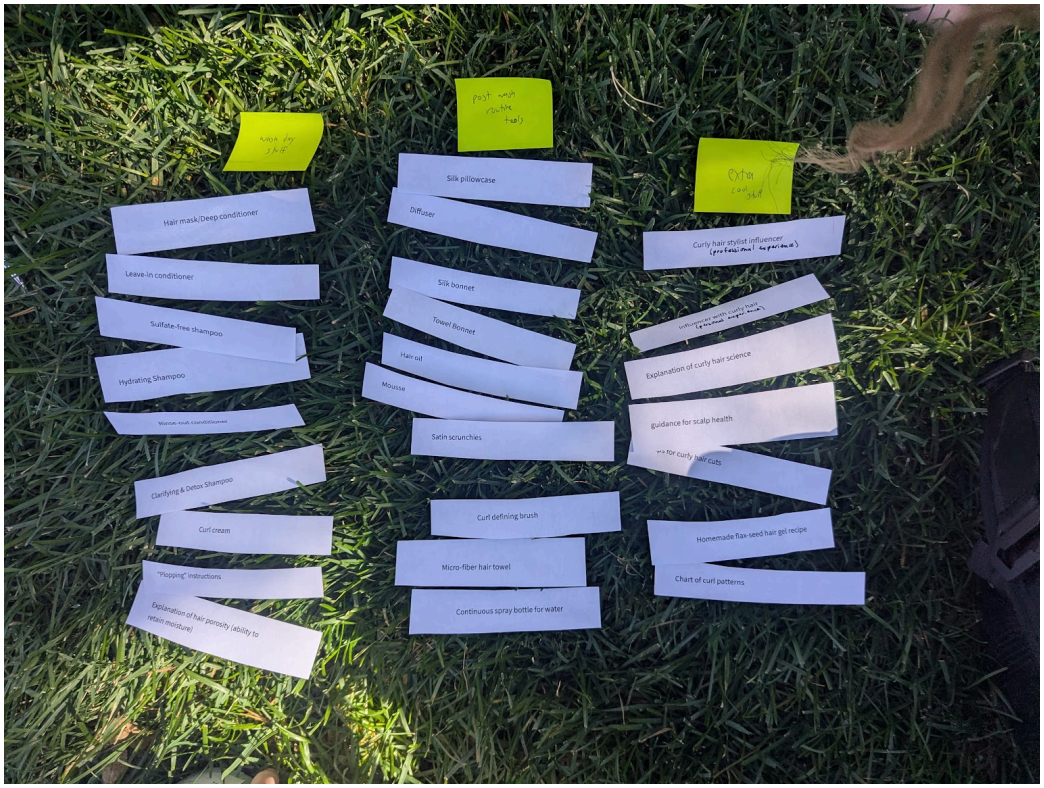
Names of participants:

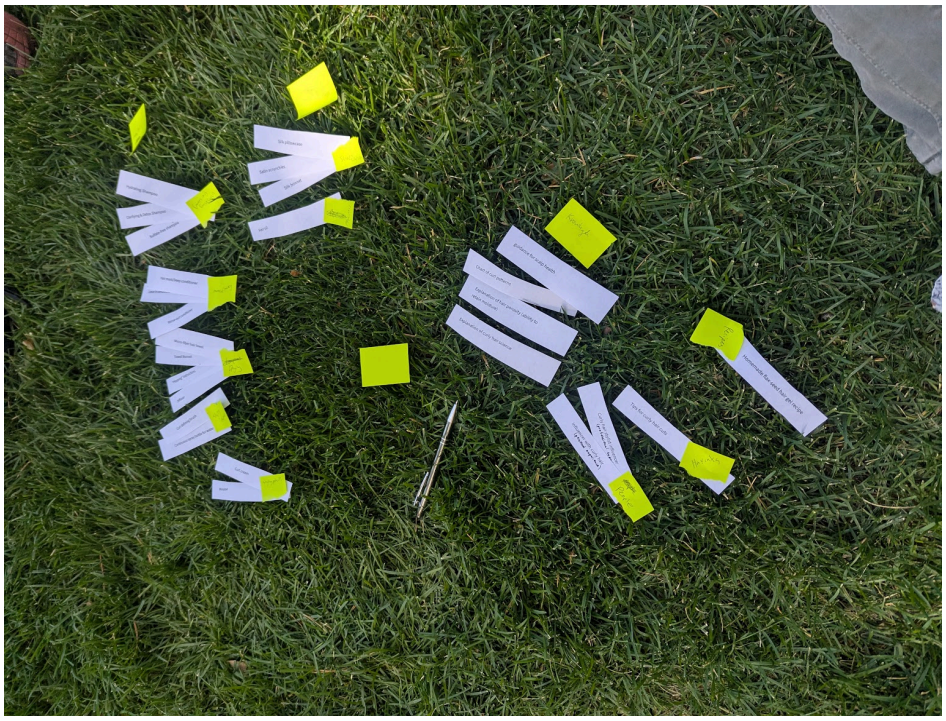
1. Amanda
2. Sarah
3. Evelyn
4. Amelia
5. Chloe
6. Lia
7. Abby
8. Beck
9. Dorene
10. Tracy

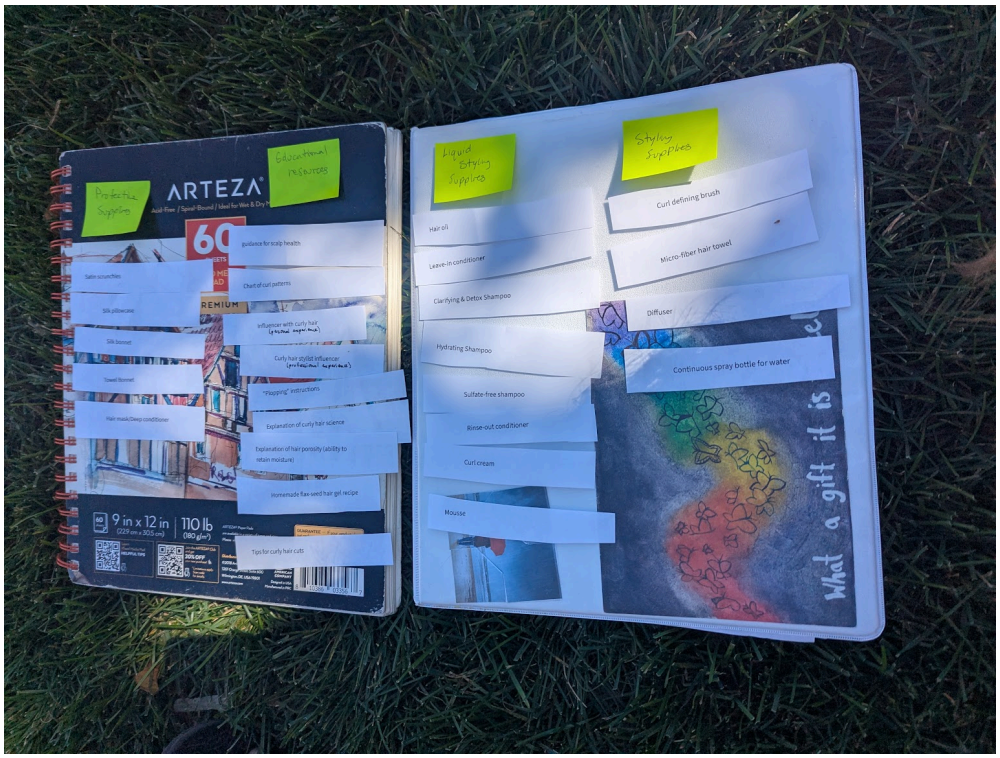
Pictures of finished card sorts:

1. 

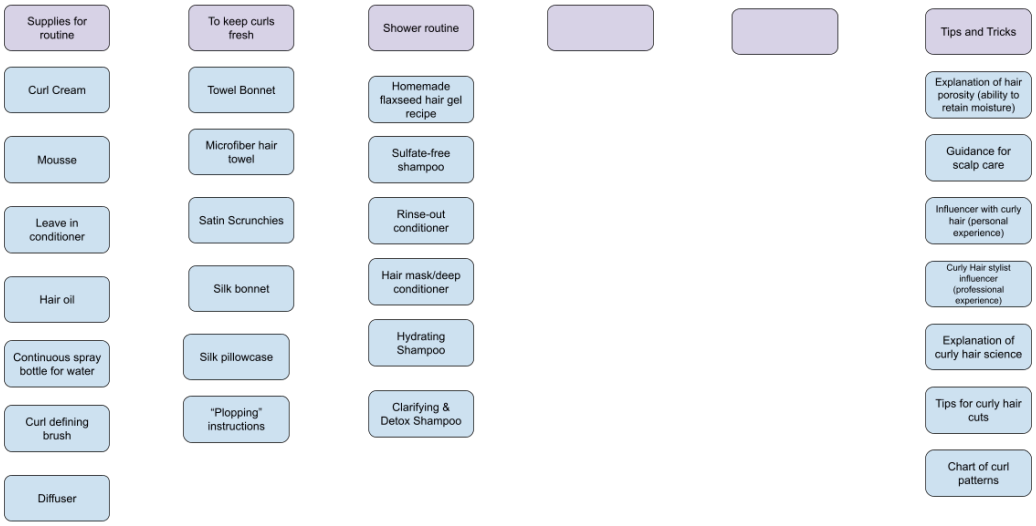
2. 
Products
Leave-in conditioner
Rinse-out conditioner
Mousse
Clarifying & Detox Shampoo
Curl cream
Hair mask/Deep conditioner
Hydrating Shampoo
Hair oil
Homemade flax-seed hair gel recipe
Sulfate-free shampoo
Items
Diffuser
Silk pillowcase
Micro-fiber hair towel
Satin scrunchies
Towel Bonnet
Curl defining brush
Continuous spray bottle for water
Silk bonnet
Science
Explanation of curly hair science
Chart of curl patterns
Guidance for scalp health
Explanation of hair porosity (ability to retain moisture)
Advice
"Plopping" instructions
Tips for curly hair cuts
Curly hair stylist influencer
Influencer with curly hair

3. 
Silk pillowcase
Diffuser
Silk bonnet
Towel Bonnet
Hair oil
Mousse
Satin scrunchies
Curl defining brush
Micro-fiber hair towel
Continuous spray bottle for water
Hair mask/Deep conditioner
Leave-in conditioner
Sulfate-free shampoo
Hydrating Shampoo
Clarifying & Detox Shampoo
Curl cream
"Plopping" instructions
Explanation of hair porosity (ability to retain moisture)
Curly hair stylist influencer
Explanation of curly hair science
guidance for scalp health
Homemade flax-seed hair gel recipe
Chart of curl patterns

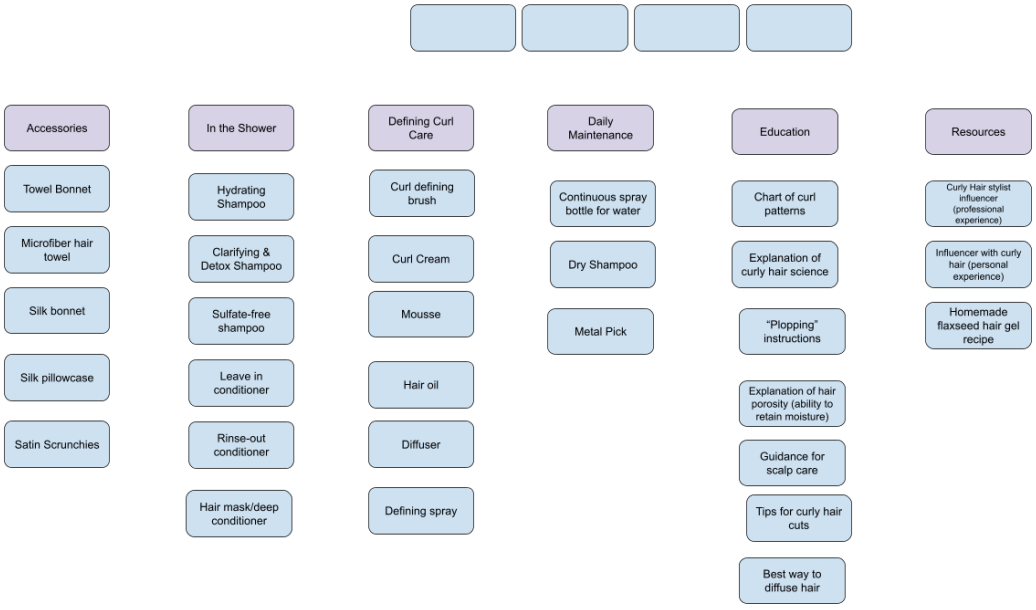
4. 

5. 
ARTEZA
Protective Supplies
Educational Resources
Liquid Styling Supplies
Styling Supplies
Hair oil
Leave-in conditioner
Clarifying & Detox Shampoo
Hydrating Shampoo
Sulfate-free shampoo
Rinse-out conditioner
Curl cream
Mousse
Curl defining brush
Micro-fiber hair towel
Diffuser
Continuous spray bottle for water
What a gift it is
9 in x 12 in
110 lb

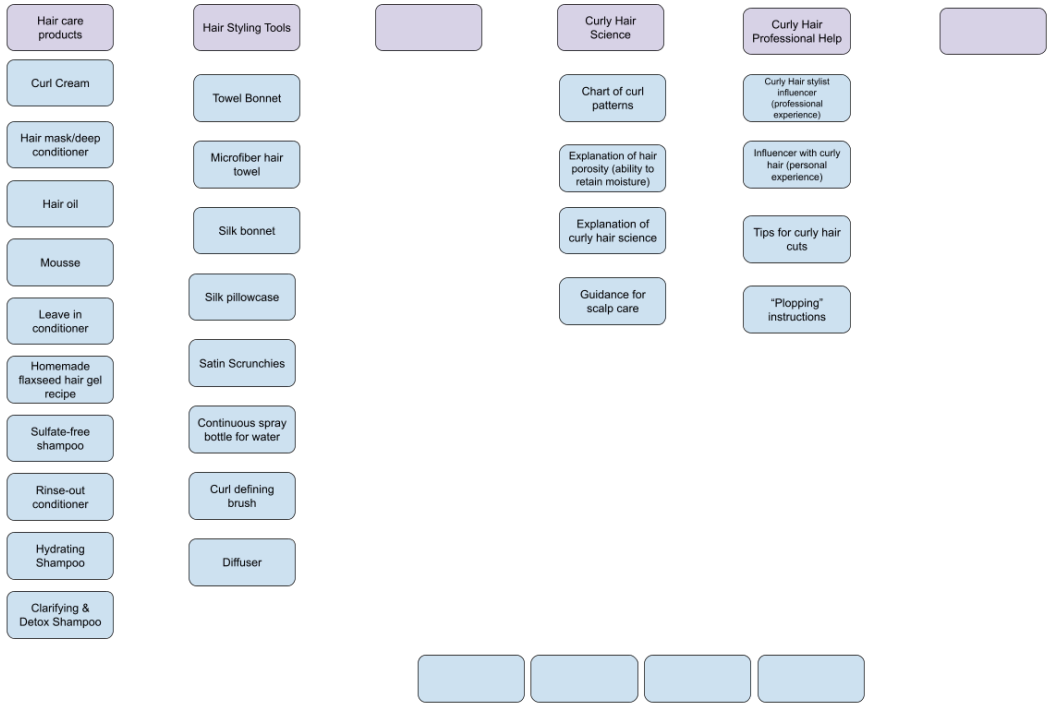
6.


Supplies for routine
Curl Cream
Mousse
Leave in conditioner
Hair oil
Continuous spray bottle for water
Curl defining brush
Diffuser
To keep curls fresh
Towel Bonnet
Microfiber hair towel
Satin Scrunchies
Silk bonnet
Silk pillowcase
"Plopping" instructions
Shower routine
Homemade flaxseed hair gel recipe
Sulfate-free shampoo
Rinse-out conditioner
Hair mask/deep conditioner
Hydrating Shampoo
Clarifying & Detox Shampoo
Tips and Tricks
Explanation of hair porosity (ability to retain moisture)
Guidance for scalp care
Influencer with curly hair (personal experience)
Curly Hair stylist influencer (professional experience)
Explanation of curly hair science
Tips for curly hair cuts
Chart of curl patterns

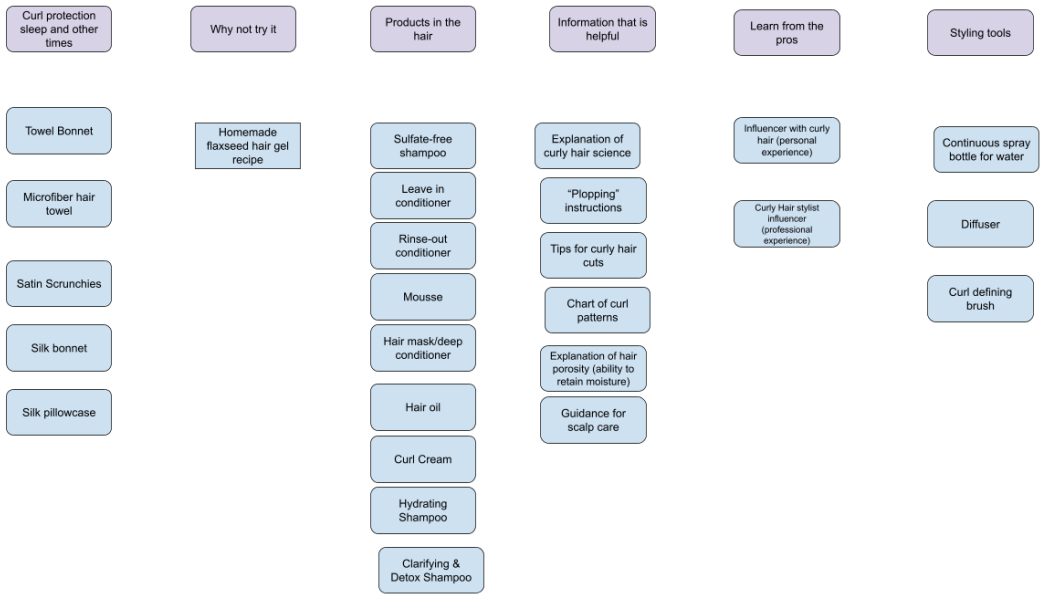
7.


Accessories
Towel Bonnet
Microfiber hair towel
Silk bonnet
Silk pillowcase
Satin Scrunchies
In the Shower
Hydrating Shampoo
Clarifying & Detox Shampoo
Sulfate-free shampoo
Leave in conditioner
Rinse-out conditioner
Hair mask/deep conditioner
Defining Curl Care
Curl defining brush
Curl Cream
Mousse
Hair oil
Diffuser
Defining spray
Daily Maintenance
Continuous spray bottle for water
Dry Shampoo
Metal Pick
Education
Chart of curl patterns
Explanation of curly hair science
"Plopping" instructions
Explanation of hair porosity (ability to retain moisture)
Guidance for scalp care
Tips for curly hair cuts
Best way to diffuse hair
Resources
Curly Hair stylist influencer (professional experience)
Influencer with curly hair (personal experience)
Homemade flaxseed hair gel recipe

8.




9.




10.

Table summarizing card sort results:

| Item Name | Participant 1 | Participant 2 | Participant 3 | Participant 4 | Participant 5 | Participant 6 | Participant 7 | Participant 8 | Participant 9 | Participant 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| **Clarifying & Detox Shampoo** | Product: shampoo | Products | wash day stuff | Washing and styling: Washing (in shower) | Liquid Styling Supplies | curly hair products | Shower routine | In the Shower | Hair care products | Products in the hair |
| **Sulfate-free shampoo** | Product: shampoo | Products | wash day stuff | Washing and styling: Washing (in shower) | Liquid Styling Supplies | curly hair products | Shower routine | In the Shower | Hair care products | Products in the hair |
| **Hydrating Shampoo** | Product: shampoo | Products | wash day stuff | Washing and styling: Washing (in shower) | Liquid Styling Supplies | curly hair products | Shower routine | In the Shower | Hair care products | Products in the hair |
| **Rinse-out conditioner** | Product: conditioner | Products | wash day stuff | Washing and styling: Conditioner | Liquid Styling Supplies | curly hair products | Shower routine | In the Shower | Hair care products | Products in the hair |
| **Hair mask / Deep conditioner** | Product: | Products | wash day stuff | Washing and styling: | Protective | curly hair | Shower routine | In the Shower | Hair care | Products in the hair |

| Item Name | Participant 1 | Participant 2 | Participant 3 | Participant 4 | Participant 5 | Participant 6 | Participant 7 | Participant 8 | Participant 9 | Participant 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| | conditioner | | | Conditioner | Supplies | products | | | products | |
| **Leave-in conditioner** | Product: after stuff | Products | wash day stuff | Washing and styling: Conditioner | Liquid Styling Supplies | curly hair products | Supplies for routine | In the Shower | Hair care products | Products in the hair |
| **Curl cream** | Product: after stuff | Products | wash day stuff | Washing and styling: Styling products | Liquid Styling Supplies | curly hair products | Supplies for routine | Defining Curl Care | Hair care products | Products in the hair |
| **Mousse** | Product: after stuff | Products | post wash routine tools | Washing and styling: Styling products | Liquid Styling Supplies | curly hair products | Supplies for routine | Defining Curl Care | Hair care products | Products in the hair |
| **Hair oil** | Product: after stuff | Products | post wash routine tools | Silk + satin: Hair oil | Liquid Styling Supplies | curly hair products | Supplies for routine | Defining Curl Care | Hair care products | Products in the hair |
| **Homemade flax-seed hair gel recipe** | Product: after stuff | Products | Extra cool stuff | Recipes | Educational resources | curly hair products | Shower routine | Resources | Hair care products | Why not try it |

| Item Name | Participant 1 | Participant 2 | Participant 3 | Participant 4 | Participant 5 | Participant 6 | Participant 7 | Participant 8 | Participant 9 | Participant 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| **Silk pillowcase** | Accessories: Put hair in/on | Items | post wash routine tools | Silk + satin: Sleep + Care | Protective Supplies | Curly hair Aftercare | To keep curls fresh | Accessories | Hair Styling Tools | Curl protection sleep and other times |
| **Satin scrunchies** | Accessories: Put hair in/on | Items | post wash routine tools | Silk + satin: Sleep + Care | Protective Supplies | Curly hair Aftercare | To keep curls fresh | Accessories | Hair Styling Tools | Curl protection sleep and other times |
| **Micro-fiber hair towel** | Accessories: Put hair in/on | Items | post wash routine tools | Washing and styling: Drying | Styling Supplies | Curly hair Aftercare | To keep curls fresh | Accessories | Hair Styling Tools | Curl protection sleep and other times |
| **Silk bonnet** | Accessories: Put hair in/on | Items | post wash routine tools | Silk + satin: Sleep + Care | Protective Supplies | Curly hair Aftercare | To keep curls fresh | Accessories | Hair Styling Tools | Curl protection sleep and other times |
| **Towel Bonnet** | Accessories: Put hair in/on | Items | post wash routine tools | Washing and styling: Drying | Protective Supplies | Curly hair Aftercare | To keep curls fresh | Accessories | Hair Styling Tools | Curl protection sleep and other times |

| Item Name | Participant 1 | Participant 2 | Participant 3 | Participant 4 | Participant 5 | Participant 6 | Participant 7 | Participant 8 | Participant 9 | Participant 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| **Diffuser** | Accessories: touch up | Items | post wash routine tools | Washing and styling: Drying | Styling Supplies | Curly hair Aftercare | Supplies for routine | Defining Curl Care | Hair Styling Tools | Styling tools |
| **Curl defining brush** | Accessories: touch up | Items | post wash routine tools | Washing and styling: Styling equipment | Styling Supplies | Curly hair Aftercare | Supplies for routine | Defining Curl Care | Hair Styling Tools | Styling tools |
| **Continuous spray bottle for water** | Accessories: touch up | Items | post wash routine tools | Washing and styling: Styling equipment | Styling Supplies | Curly hair Aftercare | Supplies for routine | Daily Maintenance | Hair Styling Tools | Styling tools |
| **"Plopping" instructions** | Resources: to-do | Advice | wash day stuff | Knowledge | Educational resources | Curly hair tips | To keep curls fresh | Education | Curly Hair Professional Help | Information that is helpful |
| **Tips for curly hair cuts** | Resources: to-do | Advice | Extra cool stuff | Hair cuts | Educational resources | Curly hair tips | Tips and Tricks | Education | Curly Hair Professional Help | Information that is helpful |
| **Guidance for scalp health** | Resources: to-do | Science | Extra cool stuff | Knowledge | Educational | Curly hair tips | Tips and Tricks | Education | Curly Hair | Information that |

| Item Name | Participant 1 | Participant 2 | Participant 3 | Participant 4 | Participant 5 | Participant 6 | Participant 7 | Participant 8 | Participant 9 | Participant 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | resources | | | | Science | is helpful |
| **Chart of curl patterns** | Resources: Science | Science | Extra cool stuff | Knowledge | Educational resources | Curly hair tips | Tips and Tricks | Education | Curly Hair Science | Information that is helpful |
| **Explanation of curly hair science** | Resources: Science | Science | Extra cool stuff | Knowledge | Educational resources | Curly hair tips | Tips and Tricks | Education | Curly Hair Science | Information that is helpful |
| **Explanation of hair porosity (ability to retain moisture)** | Resources: Science | Science | wash day stuff | Knowledge | Educational resources | Curly hair tips | Tips and Tricks | Education | Curly Hair Science | Information that is helpful |
| **Curly hair stylist influencer (professional experience)** | Resources: People | Advice | Extra cool stuff | people | Educational resources | [Separate / Uncategorized] | Tips and Tricks | Resources | Curly Hair Professional Help | Learn from the pros |
| **Influencer with curly hair (personal experience)** | Resources: People | Advice | Extra cool stuff | people | Educational resources | [Separate / Uncategorized] | Tips and Tricks | Resources | Curly Hair Professional Help | Learn from the pros |

Standardization Table

| Standardized Category | Sorts | Verbatim Pile Names |
|---|---|---|
| people | 4/10 | Resources: people (1), people (4), resources (8), Learn from the pros (10) |
| Curly Hair Products | 5/10 | Products (2), Liquid Styling Supplies (5), Curly Hair Products (6), Hair Care Products (9), Products in the Hair (10) |
| Education | 10/10 | Resources (1), Science/Advice (2), Extra cool stuff (3), Knowledge (4), Educational Resources (5), Curly hair tips (6), tips and tricks (7), Education (8), Curly hair science / curly hair professional help (9), information that is helpful (10) |
| Wash day | 3/10 | wash day stuff (2), Shower routine (7), In the Shower |
| protection | 5/10 | Accessories: put hair in / on (1), silk and satin (4), protective supplies (5), to keep curls fresh (7), Curl protection sleep and other times (10) |
| Shampoo/Washing | 2/10 | Product: Shampoo (1), Washing and Styling: Washing (in shower) (4) |
| Conditioner | 2/10 | Product: Conditioner (1), Washing and Styling: Conditioner (4) |
| Accessories/Tools | 9/10 | Accessories (1), items (2), post wash routine tools (3), Washing and styling: Styling equipment (4), styling supplies / protective supplies (5), curly hair aftercare (6), Accessories (8), hair styling tools (9), styling tools (10) |
| Styling Tools | 6/10 | Accessories: Touch Up (1), Washing and Styling: Styling equipment (4), Styling Supplies (5), Supplies for routine (7), Defining Curl Care (8), Styling Tools (10) |
| Recipes | 1/10 | Recipes (4), Why not try it (10) |
| Hair cuts | 1/10 | Hair cuts (4) |

Final categories (with all items)

Topic categories:

- Product
    - Shampoo/Conditioner
        - Sulfate-free shampoo
        - Hair mask/deep conditioner
        - Clarifying & Detox Shampoo
        - Hydrating Shampoo
        - Rinse-out conditioner
        - Leave-in conditioner
    - Styling Products
        - Mousse
        - Curl Cream
        - Hair Oil
        - Leave-in conditioner
        - Flaxseed hair gel recipe
- Tools
    - Styling
        - Diffuser
        - Curl defining brush
        - Continuous spray bottle for water
    - Protecting
        - Satin scrunchie
        - Silk bonnet
        - Silk pillowcase
        - Microfiber hair towel
        - Towel bonnet
- Education
    - Instructions
        - Flaxseed hair gel recipe
        - “Plopping” instructions
        - Tips for curly hair cuts
    - Science

        - Explanation of curly hair science
        - Chart of curl patterns
        - Explanation of hair porosity
        - Guidance for scalp health
    - Influencers
        - Curly hair stylist influencer (professional experience)
        - Influencer with curly hair (personal experience)

Sequential Labels:

- Washing
    - Clarifying & Detox Shampoo
    - Sulfate-free shampoo
    - Hydrating Shampoo
    - Rinse-out conditioner
    - Hair mask / Deep conditioner
    - Leave-in conditioner
    - Guidance for scalp health
    - Explanation of hair porosity (ability to retain moisture)
    - Chart of curl patterns
    - Curly hair stylist influencer
    - Influencer with curly hair
- Drying
    - Micro-fiber hair towel
    - Diffuser
    - Towel Bonnet
    - "Plopping" instructions
    - Curly hair stylist influencer
    - Influencer with curly hair
- Styling
    - Curl cream
    - Mousse
    - Hair oil
    - Homemade flax-seed hair gel recipe
    - Satin scrunchies
    - Curl defining brush
    - Tips for curly hair cuts

    - Chart of curl patterns
    - Curly hair stylist influencer
    - Influencer with curly hair
- Sleeping
    - Silk pillowcase
    - Silk bonnet
    - Satin scrunchies
    - Curly hair stylist influencer
    - Influencer with curly hair
- Touch up
    - Hair oil
    - Satin scrunchies
    - Continuous spray bottle for water
    - Tips for curly hair cuts
    - Chart of curl patterns
    - Curly hair stylist influencer
    - Influencer with curly hair